

Art Analysis

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1. Introduction

This project uses network science to investigate the relationships among artists, movements, institutions, and schools within the WikiArt dataset. With the rise of digitization in the cultural heritage sector, museums, galleries, libraries, and archives are increasingly placing collections online. While this change improves access, it also calls for innovative ways to engage with art beyond simply displaying physical collections on digital platforms. Effective digital engagement requires new tools to explore, connect, and interpret these resources.

The primary goal of this analysis is to answer key questions that reveal the dynamics of the art world. Specifically, this project aims to identify the most influential artists and movements, highlighting those who have significantly shaped artistic expression and development. Additionally, the analysis will explore which institutions play pivotal roles in art education and dissemination, as well as which nationalities are home to the largest concentration of artists. Finally, the project seeks to uncover the biggest communities within the network, illustrating how different entities collaborate and interact. Through exploratory data analysis, network creation, and the application of centrality measures and clustering algorithms, this project aims to provide insights into the complex web of connections that define the art landscape.

2. Methodology

This section outlines the approach used for preprocessing, exploratory data analysis (EDA), network creation, and analysis.

2.1 EDA

To understand the dataset's structure and content, I began by examining its shape, values, and distributions and identifying missing values across various attributes. Although missing values were noted, they were not handled at this stage; checks for them will be incorporated during network creation and analysis to ensure data consistency. Extensive preprocessing was not required at this initial stage.

In conducting EDA, I aimed to gain insights into the dataset as a foundation for network analysis. I first visualized the distribution of movements using a count plot (Figure 1), providing an initial perspective on movement representation across the dataset.

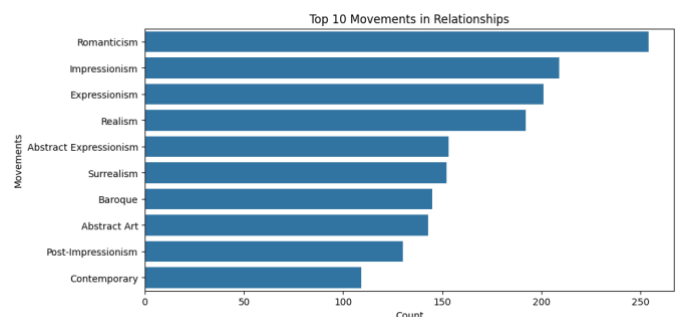


Figure 1: Movement distribution

This visualization serves as a reference for validating influential movements within the network analysis. Next, a box plot (Figure 2) of the total works by each artist revealed numerous outliers, pointing to a few artists with exceptionally high productivity. Additional graphs were generated for institutions and nationalities, as

documented in the Colab notebook.

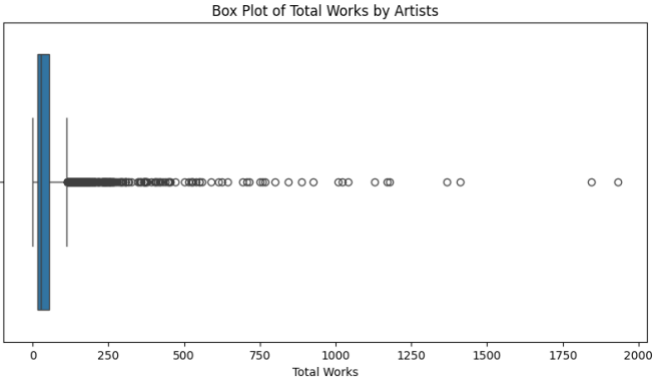


Figure 2: Boxplot for total artworks by artists

2.2 Network Creation and Analysis

To build the network, I established nodes representing significant entities such as Artists, Institutions, Schools, and Movements and connected them through edges based on their relationships. In this network, both uni-directional and bi-directional edges are present. For instance, friendships are represented as bi-directional edges since if Artist A is a friend of Artist B, then B is also a friend of A. This logic applies similarly to institutions, friendships, and movements. Initially, I created a sample network using 2% of the data (Figure 3) to confirm the structure; once validated, the full dataset was used to construct the complete network.

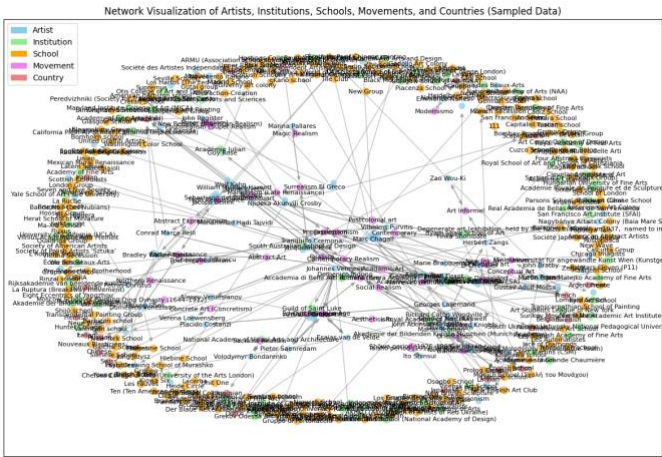


Figure 3: Network Visualization of Artists, Institutions, Schools, Movements, and Countries (Sampled Data)

Given the extensive nature of the network, I also developed sub-networks, such as those connecting only Artists and Movements, to improve visualization clarity. These sub-networks served both as visualization tools and as a way to cross-validate findings in the network analysis, described later in the report. Sub-networks include:

- Artist-School Analysis:**
A sub-network focusing on relationships between artists and schools. I visualized all

relationships initially and then filtered to show schools with over 25 artists for a clearer view of prominent institutions.

- Artist-Movement Analysis:**
This sub-network captures connections between artists and movements. The first graph represents all artists and movements, and the second filters for movements with over 200 associated artists.
- Artist-InfluencedBy Analysis:**
To reveal artist influence, I created a sub-network showing artists influenced by others. One graph displays all artists, while the second focuses on those with an out-degree greater than 6.
- Friends Analysis:**
This graph showcases relationships among artists who were friends with more than two others. By filtering nodes with a degree greater than two, the visualization emphasizes stronger connections.

2.3 Interactive Visualization

To enhance engagement with the network, I developed an interactive visualization using Plotly and Ngrok. This allows for an interactive experience, where users can explore the network’s structure dynamically.

2.4 Centrality Measures and Their Relevance

To assess the influence of various entities within the network, four centrality measures were utilized: Degree Centrality, Betweenness Centrality, PageRank, and Eigenvector Centrality. Each of these metrics provides a unique perspective on the nodes’ importance:

- Degree Centrality:**
This measure counts the number of direct connections a node has. A higher degree indicates a more connected artist, institution, or movement, suggesting greater visibility and influence within the network.
- Betweenness Centrality:**
This metric quantifies how often a node acts as a bridge along the shortest path between two other nodes. Higher betweenness indicates that the node plays a critical role in connecting disparate parts of the network, thus facilitating interactions and exchanges.
- PageRank:**
Originally developed by Google to rank web pages, this measure assesses the importance of a node based on its connections and the quality of those connections. A high PageRank signifies that an entity is not just well-connected but is also linked to other influential nodes.

4. Eigenvector Centrality:

Similar to PageRank, this measure considers the influence of a node's connections. However, it places more weight on connections to nodes that are themselves influential, providing a more nuanced view of a node's significance within the network.

These measures together allow for a comprehensive analysis of influence across different categories—artists, institutions, and movements.

3. Results

To calculate the influence metrics (Sections 3.1–3.3), I applied degree, betweenness, PageRank, and eigenvector centralities to all nodes in the graph G . This approach involved assessing each node's connections relative to others in the network, capturing different aspects of influence across artists, institutions, and movements. Using multiple metrics allowed for examining the network from various perspectives, providing a more nuanced understanding of its structure and dynamics.

Based on the application of the above methodology, the following findings emerged:

3.1. Most Influential Artists

Based on various influence measures:

- **By Degree:** Pablo Picasso, Amedeo Modigliani, Jackson Pollock, Paul Gauguin and Pierre-Auguste stand out as top influencers..
- **By Betweenness:** Key artists such as Pablo Picasso, Tintoretto, Johannes Vermeer, Camille Pissarro, and Hermann David Salomon Corrodi show high betweenness scores, indicating they play crucial brokerage roles and bridge different network segments.
- **By Pagerank:** Artists including Ancient Egypt, Titian, Pablo Picasso, Nicolas Poussin, and Caravaggio maintain high pagerank scores, which highlights their significance in overall network connectivity and influence.
- **By Eigenvector:** Artists like Edouard Manet, Edgar Degas, Fujishima Takeji, John Everett Millais, and Fyodor Bronnikov emerge as locally influential.

Summary: Pablo Picasso consistently appears as a top influencer across most of the measures, indicating broad network influence.

3.2. Most Influential Institutions

Based on various influence measures:

- **By Degree:** École des Beaux-Arts, Guild of Saint Luke, Imperial Academy of Arts, Royal Academy of Arts (RA), and Académie Julian stand out.
- **By Betweenness:** Institutions like École des Beaux-Arts, self-taught artists ,Guild of Saint Luke, Royal Academy of Arts (RA), and Kunstakademie Düsseldorf display high betweenness, acting as key hubs connecting artists across movements and institutions.
- **By Pagerank:** École des Beaux-Arts, Guild of Saint Luke, Royal Academy of Arts (RA), Imperial Academy of Arts, and self-taught artists hold high scores, indicating broad influence over a wider network of artists and schools.
- **By Eigenvector:** Institutions such as École des Beaux-Arts, Imperial Academy of Arts, Royal Academy of Arts (RA), Académie des Beaux-Arts, and Académie Julian are well-connected within clusters of other important institutions, suggesting their local influence within specific artistic circles.

Summary: École des Beaux-Arts emerges as the most influential institution across all centrality measures, serving as both a central and locally influential hub within the artist network.

3.3. Most Influential Movements

Based on various influence measures:

- **By Degree:** Movements like Romanticism, Impressionism, Expressionism, Realism, and Abstract dominate in this analysis.
- **By Betweenness:** Movements such as Expressionism, Romanticism, Impressionism, Abstract Art, and Surrealism act as bridging entities, connecting artists across different styles.
- **By Pagerank:** Romanticism, Realism, Impressionism, Expressionism, and Surrealism rank highly, indicating substantial overall influence and significance within the network.
- **By Eigenvector:** Movements like Romanticism, Impressionism, Realism, Academic Art, and

Expressionism Surrealism emerge as central within their local clusters.

Summary: Romanticism and Impressionism are consistently influential across various centrality measures, indicating broad influence and central roles in connecting artists and institutions within the network.

3.4. Concentration of nationalities

American and French artists dominate the network, followed by Italian and British artists, with each nationality contributing distinct cultural and artistic influences across various movements. This distribution can be seen in Figure 4.

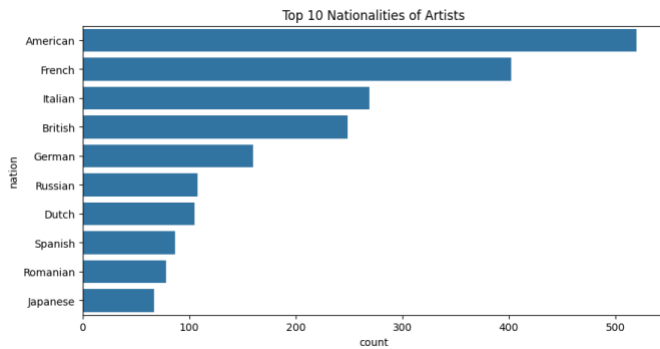


Figure 4: Distribution of Artist Nationalities

3.5. Communities in Network

In the artist network, community detection revealed five significant clusters, with the largest comprising 132 artists. The next largest communities include clusters of 93, 76, 74, and 49 artists. These groupings suggest distinct sub-networks where artists are closely connected, potentially due to shared influences, movements, or regions. The largest community visualized (Figure 5) also indicates a tight-knit structure, underscoring the presence of influential clusters within the broader network. This insight into community structure helps identify core artist groups that may have had substantial mutual influence or shared artistic heritage.

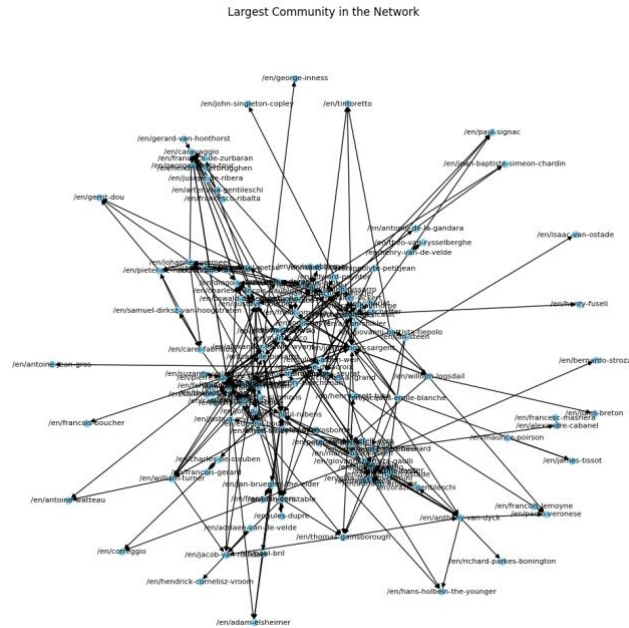


Figure 5: Largest Community in the Network.

4. Conclusion

The exploratory data analysis (EDA) phase was manageable due to prior experience, enabling quick insights into data structure and distribution. However, the network analysis presented a significant challenge, as it was my first attempt at applying these methods. Building the final network took multiple iterations, requiring adjustments to address initial inconsistencies in network construction. Despite these hurdles, this learning experience was rewarding, providing deeper insights into network dynamics and the influence structures within the dataset.

Visualization was also crucial due to the network's complexity, with many nodes and edges. Using a divide-and-conquer approach helped focus on smaller sections of the network, making analysis clearer and more organized. Early visualizations aligned with the network analysis results, which helped confirm and cross-check my initial findings.

In future work, addressing the missing values in the relationship data could further enhance the network's accuracy and depth. Additionally, this dataset holds potential for predictive analysis on the evolution of artistic movements, allowing us to explore how combinations of existing movements might influence future art trends.